Systems & Inequalities

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.01

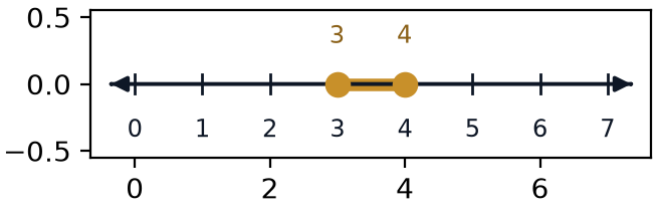
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A geometric figure scaled by factor $k = 4$ has an original side length of 4 cm. What is the corresponding image length? Show your complete algebraic work.

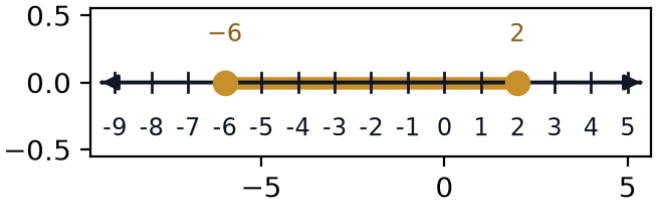
2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 4)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.



3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

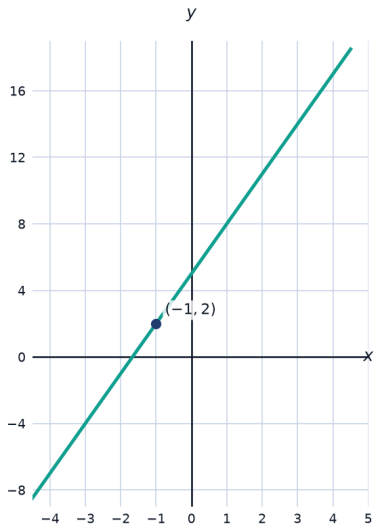
In ΔABC studying geometric sequences as exponential functions, $m\angle A = 30^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$? Show your complete algebraic work.



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4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 3x + 5$ in comparing exponential relationships, evaluate $f(2)$.

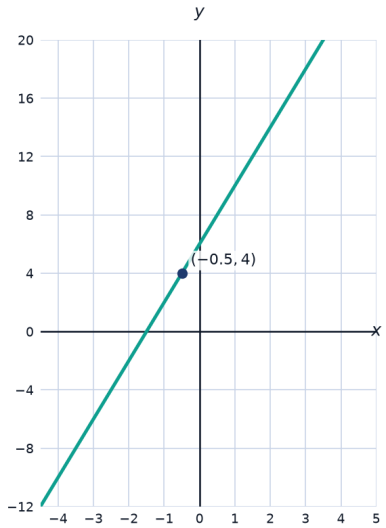


5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for comparing exponential relationships consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 4x + 6$ in comparing linear and exponential relationships, evaluate $f(2)$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

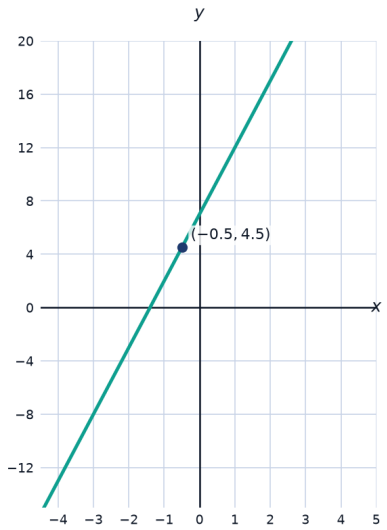
A data set recorded for comparing linear and exponential relationships consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean? Show your complete algebraic work.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 5x + 7$ in simple interest, evaluate $f(2)$.



9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Solve the linear system using matrix inversion or Cramer's Rule: $3x - 2y = 8$ $5x + 4y = 6$

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Level 3 · AdvancedIn the system of equations below, k is a constant. For what value of k does the system have no solution? $\{kx - 6y = 15; 4x - 3y = 9$

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Level 4 · SynthesisIn the system of equations $ax + 8y = 24$ and $6x + by = 8$, a and b are constants. If the system has infinitely many solutions, what is the value of $a + b$?

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Level 2 · MediumA theater sold a total of 250 adult and student tickets for an evening performance, collecting $2,400. If adult tickets cost 12 each and student tickets cost 8 each, how many student tickets were sold?

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Level 1 · FoundationsIf $5(2x - 3) = 3(x + 4) + 6$, what is the value of $7x - 15$?

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Matrices & Systems · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

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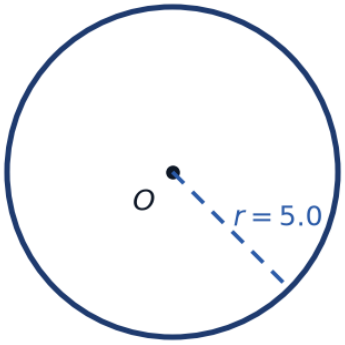
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 10.01 Rational Expressions

Find the value(s) of k for which $A = (k, 2; 3, k)$ is singular.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 11.01 Statistics & Data Analysis

The equation of a circle in the xy-plane is $(x - 5)^2 + (y + 2)^2 = 81$. What is the radius of the circle?



17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 Quadratic Functions

A solid metal cone with base radius 4 cm and height 9 cm is submerged into a cylindrical beaker with internal radius 6 cm partially filled with water. Assuming the cone is fully submerged and no water overflows, by how many centimeters does the water level in the beaker rise?

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Exponents & Polynomials

Solve the rational equation for x: $\frac{x}{x-2} + \frac{1}{x+3} = \frac{5}{x^2+x-6}$. Check for extraneous solutions.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Functions & Transformations

A committee of 4 members is to be selected at random from a club of 7 seniors and 5 juniors. (a) What is the total number of possible committees? (b) Find the probability that the committee contains exactly 2 seniors and 2 juniors.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

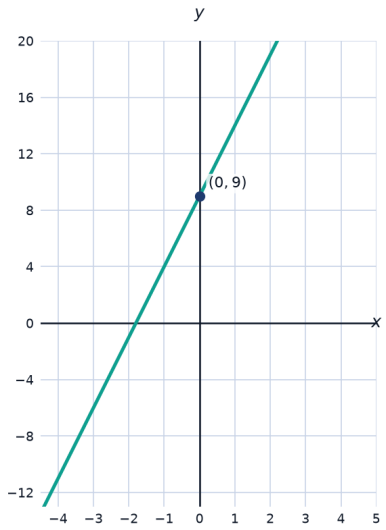
Find all real solutions to the equation $|2x - 7| = 3x - 1$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 5x + 9$ in graphs and characteristics of exponential functions, evaluate $f(2)$.



a. 19

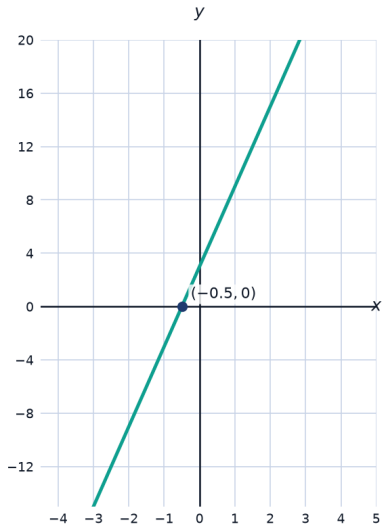
b. 24

c. 10

d. 23

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 6x + 3$ in exponential growth and decay, evaluate $f(2)$.



a. 15

b. 21

c. 12

d. 19

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for graphs and characteristics of exponential functions consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for exponential growth and decay consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5

d. 22.5

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for simple interest consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5

d. 26.5

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For eigenvalue λ with eigenvector v, any scalar multiple cv is also an eigenvector. True or False?

a. Only if $c = 1$

b. False

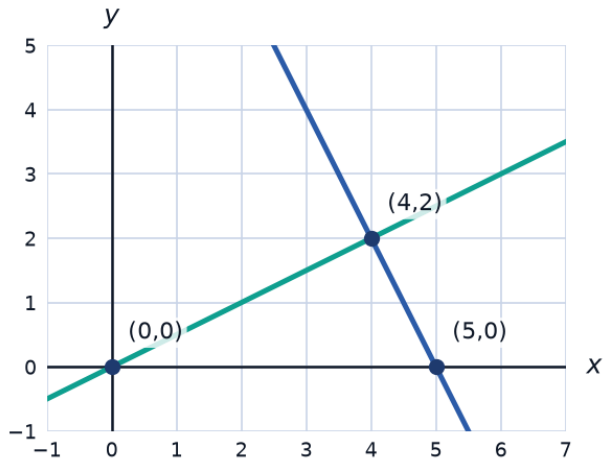
c. Only if $c > 0$

d. True

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27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the xy-plane, what is the area of the triangular region bounded by the lines $y = \frac{1}{2}x$, $y = -2x + 10$, and the x-axis?



a. 4

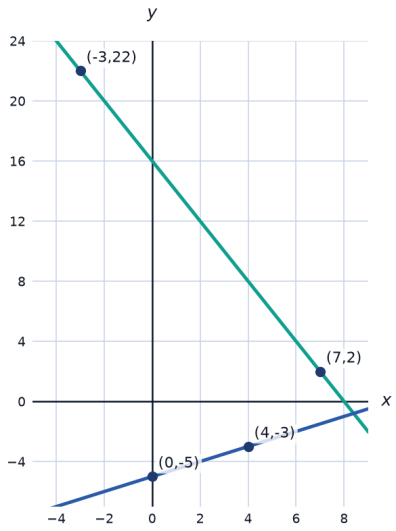
b. 5

c. 6

d. 10

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Line k passes through the points $(7, 2)$ and $(-3, 22)$. Line p is perpendicular to line k and passes through $(4, -3)$. What is the y-intercept of line p?



a. -5

b. 5

c. $-\frac{7}{2}$

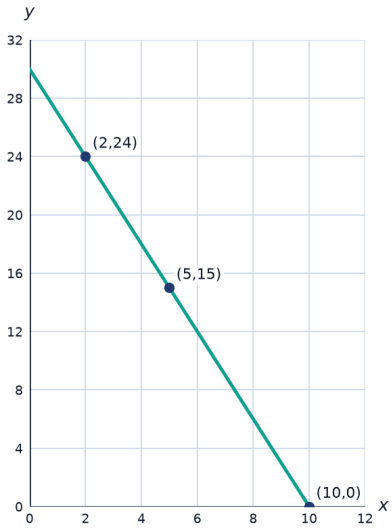
d. -2

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

The graph shows the volume of water $V(t)$, in thousands of gallons, remaining in a municipal reservoir t hours after draining begins at a constant rate. According to the linear model, after how many hours will the reservoir be completely empty?



a. 8

b. 10

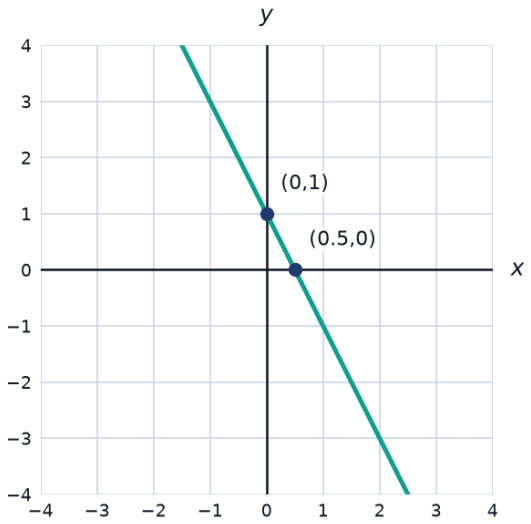
c. 12

d. 15

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30 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

The line $y = kx + 7$ intersects the line $y = -2x + 1$ at a point with a positive x-coordinate and a positive y-coordinate. Which condition must k satisfy?



a. $k > -2$

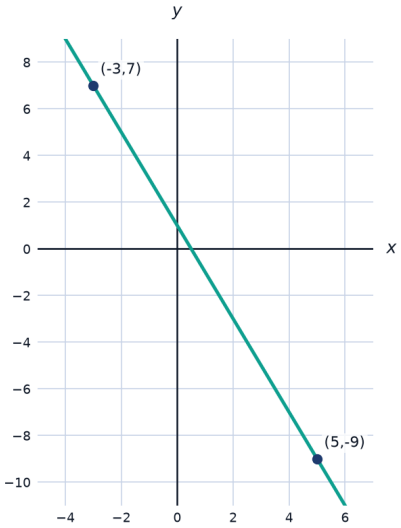
b. $k < -14$

c. $k > -7$

d. $k < -2$

31 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Line k passes through $(-3, 7)$ and $(5, -9)$. What is the x-intercept of line k?



a. $(0.5, 0)$

b. $(1, 0)$

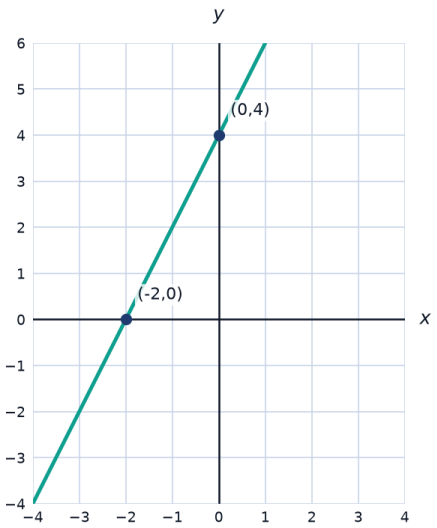
c. $(0, 1)$

d. $(2, 0)$

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32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

What is the distance between the parallel lines $y = 2x + 4$ and $y = 2x - 6$?



a. $2\sqrt{5}$

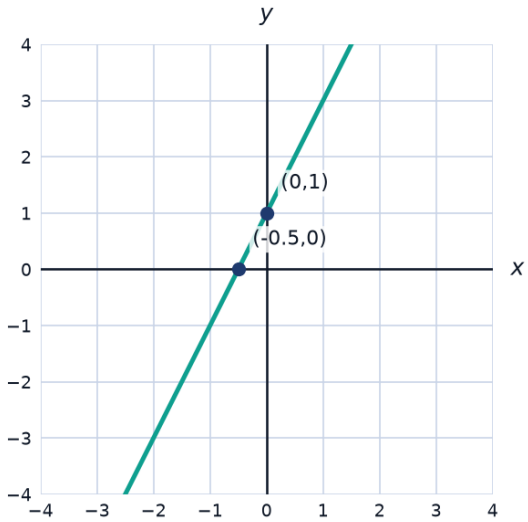
b. $\sqrt{10}$

c. 5

d. $2\sqrt{10}$

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For what value of b does the line $y = 3x + b$ pass through the intersection point of $y = 2x + 1$ and $y = -x + 10$?



a. -4

b. -2

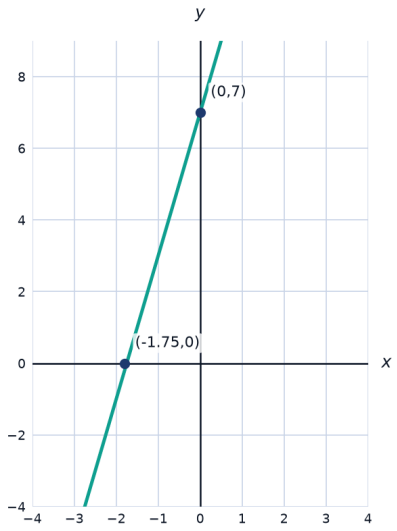
c. 2

d. 4

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A linear function has $g(0) = -3$ and its graph is perpendicular to the line $y = 4x + 7$. What is $g(12)$?



a. -6

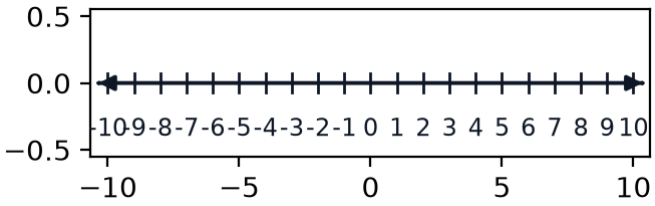
b. -4

c. 0

d. 3

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Which values of x satisfy $-2x + 3 > 7$?



a. $x > -2$

b. $x < -2$

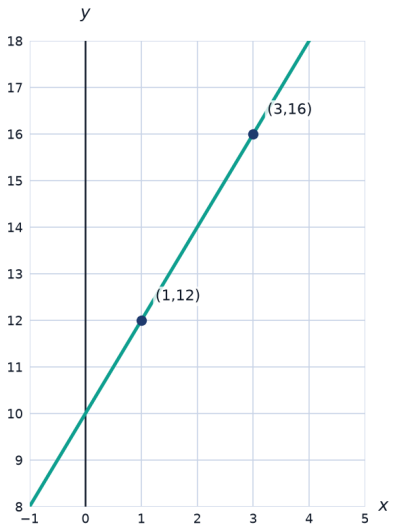
c. $x > 2$

d. $x < 2$

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A line passes through the points $(1, 12)$ and $(3, 16)$. What is its x-intercept?



a. 3

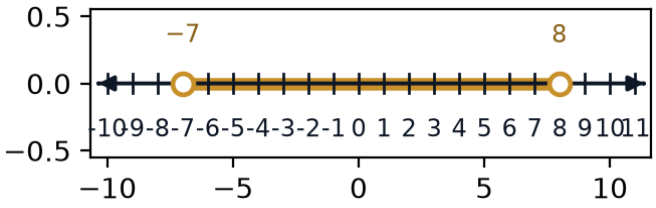
b. -11

c. -12

d. -5

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Which value of x satisfies the inequality $-3x + 8 < -7$?



a. 3

b. 4

c. 5

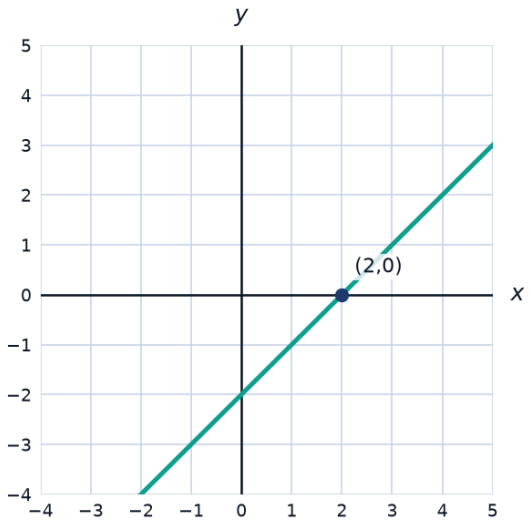
d. 6

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Exponents & Polynomials

The graph of a function is shown. At which x-value does the function equal zero?



a. $x = -2$

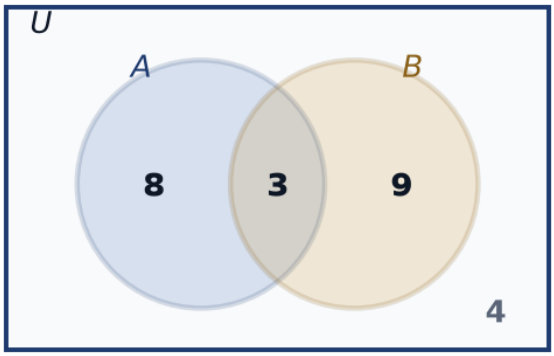
b. $x = 0$

c. $x = 1$

d. $x = 2$

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Functions & Transformations

Universal set $\mathscr{E} = \{x : x \in \mathbb{Z}^+, x \leq 15\}$. Set $A = \{x : x \text{ is a multiple of } 3\}$ and $B = \{x : x \text{ is a multiple of } 5\}$. Find $A \cap B$.



a. $\{15\}$

b. $\{3, 5, 15\}$

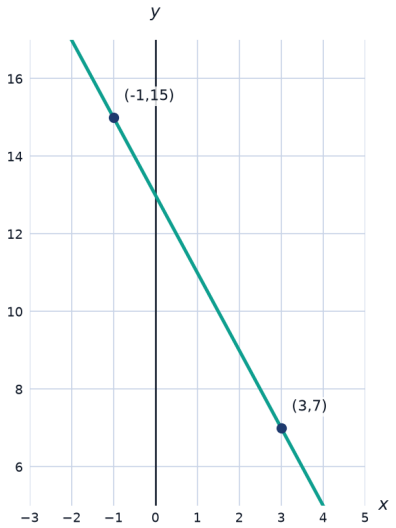
c. $\{3, 5, 6, 9, 10, 12, 15\}$

d. $\varnothing$

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

A line passes through the points $(-1, 15)$ and $(3, 7)$. If $f(x)$ represents this line, what is $f(-6)$?



a. 29

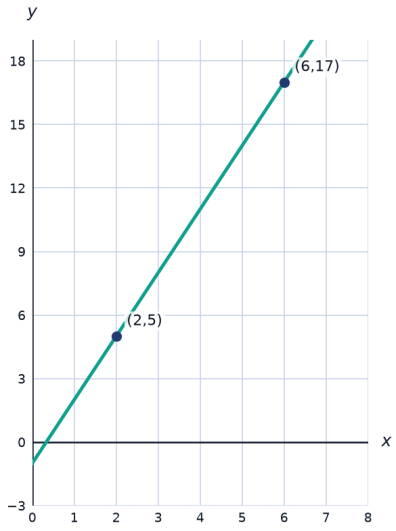
b. 16

c. 25

d. 18

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Sequences & Series

What is the slope of the line passing through $(2, 5)$ and $(6, 17)$?



a. 2

b. 3

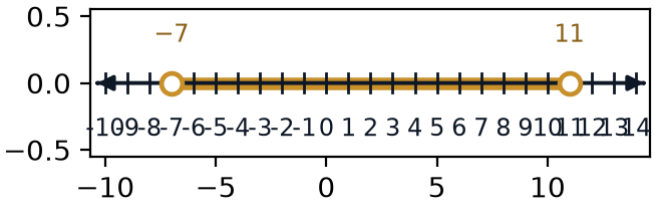
c. 4

d. 5

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 8.01 Factoring Polynomials

Which value of x satisfies $3x - 7 > 11$?



a. 4

b. 5

c. 6

d. 7

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 9.01 Radicals & Exponents

Consider the matrix equation $(2, 1, -2; 4, -1, -1; 2, -2, 1)(x; y; z) = (9; 9; k)$ where k is a real constant. Given that the system has infinitely many real solutions (x, y, z), what is the minimum possible value of $x^2 + y^2 + z^2$?

a. 9

b. 3

c. 6

d. 45

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Linear Equations & Lines

Consider the system of linear equations represented by the matrix equation $(1, 1, 1; 1, 2, 3; 3, 5, k)(x; y; z) = (6; 14; m)$ where k and m are real constants. Suppose the system is consistent and has infinitely many solutions. Let $P(x, y, z)$ be the unique solution that minimizes the squared Euclidean distance from the origin, $d^2 = x^2 + y^2 + z^2$. What is the value of $k + m + d^2$?

a. 55

b. 47

c. 38

d. 109

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 10.01 Rational Expressions

A local government is planning to build a new community park. The park will feature a rectangular garden area and a circular fountain. The garden's length, L, and width, W, are constrained by a perimeter of at most 200 meters. Specifically, $2L + 2W \leq 200$. Additionally, to ensure accessibility, the width must be at least 10 meters less than the length, so $W \geq L - 10$. The area of the garden is given by $A = LW$. The city wants to maximize the garden's area, A, subject to these constraints. If the length of the garden, L, is restricted to be between 30 and 70 meters inclusive, what is the maximum possible area of the garden?

a. 2100 square meters

b. 2400 square meters

c. 2500 square meters

d. 2600 square meters

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